

Security Linter Report

Filename	case6_augassign_conditional.py
Timestamp	2026-05-04 19:12:30 UTC
Lines of Code (LOC)	4
Verdict	VULNERABLE

ID	Severity	Sink	Source Variable	Source Type	Line
CVE-SL-002	WARNING	db.execute	user_filter	EXTERNAL	4

CRITICAL	0
WARNING	1
INFO	0
TOTAL	1

CVE-SL-002 — db.execute**WARNING**

CWE	CWE-89
Sink	db.execute
Sink Line	4
Sink Column	0
Sink Expression	db.execute(base_query)
Argument	base_query
Source Variable	user_filter
Source Type	EXTERNAL
Propagation Mechanism	data flow propagation

Taint Path Trace:

Step	Role	Line	Expression
1	SOURCE	2	if user_filter:
2	ASSIGN	3	base_query += " AND user = " + user_input + ""
3	SINK	4	db.execute(base_query)

Remediation:

Identify the data source and use a parameterized query. Never concatenate external data into SQL strings:

```
# Replace string building with a parameterized query:  
cursor.execute("SELECT * FROM t WHERE col = %s", (external_value,))
```

No safe sink paths confirmed.

No sanitization functions detected.

CFG — Total Nodes	7
CFG — Condition/Loop Nodes	1
CFG — Assignment Nodes	2
CFG — Call Nodes	1
CFG — Merge Nodes	1
██	██████████████
DFG — Total Nodes	13
DFG — Total Edges	11
DFG — Source Nodes	0
DFG — Sink Nodes	1
DFG — Operator Nodes	3
DFG — Constant Nodes	3
DFG — Variable Nodes	6
Taint Propagation Hops	3
██	██████████████
Analysis Time (ms)	14.2

Vulnerabilities	0 CRITICAL 1 WARNING 0 INFO
Safe Sinks	0
Analysis Time	14.2 ms
Engine Version	1.0.0
Final Verdict	VULNERABLE